OSCAR DAVIS

oscar.davis@cs.ox.ac.uk ◇ olsdavis.github.io

Research statement. I develop generative models based on flow matching and diffusion, with a focus on continuous flows for language modelling and few-step sampling via flow maps.

EDUCATION

PhD in Computer Science, University of Oxford — Oct 2023 – (Expected) Dec 2026
- Supervised by M. Bronstein, İ. Ceylan, J. Bose.

MSc in Advanced Computer Science, University of Oxford — Oct 2022 – Aug 2023
- Supervised by M. Bronstein and İ. Ceylan. Distinction. Best dissertation prize (see below).

Visiting Student, Imperial College London — Sep 2021 – Jul 2022
- Supervised by A. Gervais. Funded by Swiss scholarship. Distinction.

BSc in Computer Science, EPFL — Sep 2019 – Jul 2022

RESEARCH EXPERIENCE

Research Intern at Apple MLR, with M. Cuturi — Mar 2026 – Present
- Scaling Categorical Flow Maps to 1.7B parameters, 2.1T tokens. First continuous flow model of that scale for language modelling. Scaled self-distillation objectives, enabling coherent sampling with as few as 4 NFEs.
- Designing new variants of continuous flow models for language modelling.
- Pre-training discrete diffusion models for reasoning tasks.

Research Intern at Genesis Molecular AI, with J. Bose, N. Boffi, M. Al-Shedivat — Jan – Mar 2026
- Flow maps, steering and tilting for protein generation.

Research Intern at Microsoft Research, Cambridge, with J. Gladrow & K. Kalinin — Nov 2023 – Feb 2024
- Engineering diffusion models, latent diffusion models, VAEs, simple video models, neural ODEs.
- Theoretical analyses of Diffusion Models via SDEs, PDEs and fixed-points. *(Patented.)*

RESEARCH PROJECTS

MSc Dissertation, Information Theory for GNNs, with İ. Ceylan, M. Bronstein — Feb – Aug 2023
- Developed a formal information-theoretic framework to fully characterise informational bottlenecks in Graph Neural Networks, including over-smoothing and over-squashing. **Received the Tony Hoare Prize for the best dissertation of the year.**

BSc Research Project, DeFi analysis, with A. Gervais — Jan – Aug 2022
- Analysed DeFi markets on Ethereum and BNB Chain, quantifying offered financial security; implemented a custom CUDA version of the Bellman-Ford algorithm to detect arbitrage opportunities 864× faster than previous SOTA.

Student Research Project, Scala 3.0 Compiler Extension, with M. Odersky — Jun – Sep 2021
- Contributed to the thread-safe re-implementation of `lazy vals`, in the Scala 3 compiler.

PUBLICATIONS

Scaling Categorical Flow Maps — May 2026
Davis, O., Filippova, A., Ablin, P., Turrisi, V., Shidani, A., Cuturi, M., Béthune, L.
Pre-print. arXiv: arxiv.org/abs/2605.07820.

Categorical Flow Maps Jan 2026
Roos, D.*, **Davis, O.***, Eijkelboom, F.*, Bronstein, M., Welling, M., Ceylan, I., Ambrogioni, L., vd Meent, J.W.
ICML 2026. arXiv: arxiv.org/abs/2602.12233. GitHub: github.com/olsdavis/semicat.

Generalised Flow Maps for Few-Step Generative Modelling on Riemannian Manifolds Sep 2025
Davis, O., Boffi, N., Albergo, M., Bronstein, M., Bose, J.
ICLR 2026. NeurIPS 2025 FPI. arXiv: arxiv.org/abs/2510.21608. GitHub: github.com/olsdavis/gfm.

SOAPIA: Siamese-Guided Generation of Off Target-Avoiding Protein Interactions [...] May 2025
Vincoff, S.*, **Davis, O.***, Tong, A., Bose, J., Chatterjee, P.
ICML 2025 FM4LS. OpenReview: openreview.net/pdf?id=Ax25SLIDsN.

FORT: Forward-Only Regression Training of Normalizing Flows May 2025
Rehman, D., **Davis, O.**, Lu, J., Tang, J., Bronstein, M., Bengio, Y., Tong, A., Bose, J.
ICLR 2026. ICML 2025 GenBio (**Best paper award**). arXiv: arxiv.org/abs/2506.01158

SOAPI: Siamese-guided Generation of Off-Target-Avoiding Protein Interactions Mar 2025
Vincoff, S., **Davis, O.**, Tong, A., Bose, J., Chatterjee, P.
ICLR 2025 GEM (**Spotlight**). OpenReview: openreview.net/pdf?id=aRrXs2cVdy.

Fisher Flow Matching for Generative Modelling over Discrete Data May 2024
Davis, O., Kessler, S., Petrache, M., Ceylan, I., Bronstein, M., Bose, J.
NeurIPS 2024. arXiv: arxiv.org/abs/2405.14664. GitHub: github.com/olsdavis/fisher-flow.

TEACHING EXPERIENCE

Co-Lead Teaching Assistant for generative modelling at EEML 2025, Sarajevo Jul 2025
- Writing and presenting a geometric generative modelling tutorial (flow matching, Riemannian flow matching).

Graduate Teaching and Research Scholarship in CS, Oriel College, Oxford Apr 2025 – Mar 2026
- Teaching undergraduate-level courses to students of Oriel College. Admissions interviews.

TA for Geometric Deep Learning, University of Oxford, under M. Bronstein Jan – Mar 2025 and 2026
- Teaching PyTorch implementations of geometric models (equi-/invariance) and others (*e.g.,* neural diffusion).

OÄW Winter AI School 2025, Vienna Jan 2025
- Gave two PyTorch tutorials: one on implementing Graph Neural Networks; one on (Riemannian) flow matching.

TA for Graph Representation Learning, University of Oxford, under İ Ceylan Oct – Dec 2023 and 2024
- Teaching PyTorch and PyTorch Geometric (for Graph Neural Networks, and Knowledge Graph Learning).

TA for Object-Oriented Programming (Java), EPFL, under M. Schinz Feb – Jun 2020

AWARDS & SERVICE

ICML 2025 GenBio – Best paper award May 2025

NeurIPS Top Reviewer 2025

Tony Hoare Prize for the best MSc Dissertation, University of Oxford Sep 2023

Swiss Study Foundation Scholarship (for near 100% GPA in my final term) Sep 2021

G-Research Grant for PhD Students and Postdocs (£1k) Feb 2024

SKILLS

Programming Python, PyTorch, JAX (Equinox, Grain, Orbax), Java, Go, Scala
Systems Git, Linux, multi-GPU and multi-node distributed training